

EC2010

Macroeconomics: Money and Monetary Policy

Summer 2024 Examination · Paper 1
Duration 1.5 hours · Section A: 4 of 6 short questions (50%) · Section B: 1 of 2 long questions (50%)

Module	EC2010 — Macroeconomics: Money and Monetary Policy
Session	Summer 2024
Structure	Section A: six short questions, answer any four (50% of the paper between them). Section B: two long questions, answer one (50%) — Question B2 itself offers a further choice between part (b) and part (c) after the compulsory part (a).
This scheme covers	All six Section A questions, both Section B questions, and both the (b) and (c) alternatives within B2, so you can check your working whichever combination you sat.

Own-figure rule. If you carry an early error into a later part of a question but apply the right method consistently from that point on, you still pick up the marks for the later steps done correctly on your own figures.

How this paper relates to Summer 2025. Several questions here (A2, A4, A5, A6, B1(b), B2(a)) reappear in similar form on the Summer 2025 paper, sometimes with the direction of the shock reversed (e.g. a rise vs. a fall in G, a positive vs. negative energy shock). Comparing the two schemes side by side is a good way to check you understand the mechanism rather than having memorised one direction.

Section A · Question 1

SME Lending, Repayments & the Money Supply · 2 × 50 marks

A1100 marks

Explain how each of the following would be expected to affect the aggregate money supply in Ireland.

(i) Gross new lending advanced to SMEs by Irish Banks was €1.1 billion during Q3 2023.

(ii) Repayments by all SMEs to Irish Banks were €968 million in Q3 2023.

In your answer you should show how each affects the assets and liabilities of a typical commercial bank.

Source: Central Bank of Ireland Statistical Release on Trends in SME and Large Enterprise Credit and Deposits: Q3 2023.

FULL MODEL ANSWER

Both parts turn on the same mechanism: when a commercial bank makes a new loan, it doesn't hand over pre-existing money — it creates a new deposit for the borrower at the moment the loan is drawn down. That new deposit is itself part of the money supply. Repayment runs the same mechanism in reverse and destroys money.

(i) New lending of €1.1bn

The bank grants the loan and credits the SME's current account with the funds — a brand-new asset (the loan) and a brand-new liability (the deposit) are created simultaneously.

Commercial bank balance sheet	Assets	Liabilities
Loans to SMEs	+€1.1bn	—
SME current account deposits	—	+€1.1bn
Net effect	+€1.1bn	+€1.1bn

This new deposit is part of M1/M3 (broad money) and did not exist before, so the aggregate money supply **increases by €1.1bn**.

(ii) Repayments of €968m

When SMEs repay, their current account is debited and the corresponding loan asset is written down by the same amount — the deposit that funded the repayment is extinguished, not transferred to the bank as cash.

Commercial bank balance sheet	Assets	Liabilities
Loans to SMEs	–€968m	—
SME current account deposits	—	–€968m
Net effect	–€968m	–€968m

This destroys €968m of deposits, so the aggregate money supply **falls by €968m** from this flow alone.

Worth noting: here new lending (€1.1bn) exceeded repayments (€968m), so the net effect of these two flows together was a **€132m expansion** in credit outstanding to the SME sector — the opposite sign to the equivalent question on the Summer 2025 paper, where repayments exceeded new lending. Always compute the net figure explicitly rather than just stating both directions separately; examiners reward the comparison.

Mark Allocation — Part (i): New Lending (50 marks)

Tier	Marks	What separates this tier
40-50	Correct direction (increase) with a fully correct, correctly-labelled balance sheet entry showing both the asset (loan) and liability (deposit) side, and a clear statement that a new deposit is money.	
25-39	Correct direction and mostly correct balance sheet entries, but the explanation of why the deposit counts as new money is thin.	
10-24	States the loan increases the money supply with little supporting balance sheet mechanics.	
0-9	Incorrect direction, or confuses this with a simple transfer of existing funds.	

Mark Allocation — Part (ii): Repayments (50 marks)

Tier	Marks	What separates this tier
40-50	Correct direction (decrease) with fully correct balance sheet entries on both sides and a clear statement that repayment destroys deposit money.	
25-39	Correct direction, entries mostly correct but explanation of the money-destruction mechanism underdeveloped.	
10-24	States money supply falls with little supporting mechanics.	
0-9	Incorrect direction or no meaningful balance sheet treatment.	

Question A1 Total: 100 Marks

Section A · Question 2

ECB Policy Rates & Communication · 30 + 70 marks

- A2(a)

30 marks

The following relates to the ECB’s Monetary Policy Decisions, Press Release, Jan. 25th 2024: “The interest rate on the main refinancing operations and the interest rates on the marginal lending facility and the deposit facility will remain unchanged at 4.50%, 4.75% and 4.00% respectively.” Explain the deposit facility.
- A2(b)

70 marks

The following is taken from the ECB’s Monetary Policy Statement, Press Conference, Jan. 25th 2024: “We will continue to follow a data-dependent approach to determining the appropriate level and duration of restriction. In particular, our interest rate decisions will be based on our assessment of the inflation outlook in light of the incoming economic and financial data, the dynamics of underlying inflation and the strength of monetary policy transmission.” Explain how this statement differs from Forward Guidance.

FULL MODEL ANSWER

(a) The deposit facility — 30 marks

- What it is:** One of the ECB's two standing facilities (alongside the marginal lending facility). Eligible commercial banks can deposit surplus overnight liquidity directly with their national central bank on their own initiative, without needing a counterparty.
- What it does:** It sets the floor of the interest rate corridor. No bank will lend to another bank overnight at a rate below what it can earn risk-free from its own central bank, so the deposit rate anchors the bottom of short-term money market rates.
- Why the spread here is wider than usual:** note the gap this time: MRO (4.50%) sits 50bp above the deposit facility (4.00%), not the tighter 25bp gap seen in earlier years. The Governing Council deliberately widened the MRO–deposit facility spread to 50bp from 27 July 2022, specifically to reduce the incentive for banks holding cheap TLTRO III funding to simply redeposit it at the (now positive) deposit rate for a risk-free profit. This decision remained in place at the January 2024 meeting referenced here — it was only narrowed back to 15bp from September 2024.

Rate (Jan 2024)	Level	Role
Marginal lending facility	4.75%	Ceiling — emergency overnight borrowing from the Eurosystem
Main refinancing operations (MRO)	4.50%	Reference/signalling rate on regular liquidity operations
Deposit facility	4.00%	Floor — rate earned on surplus reserves parked overnight

(b) “Data-dependent, meeting-by-meeting” vs. Forward Guidance — 70 marks

Forward guidance is a commitment about the future path of policy, made explicit in advance — either time-contingent (“rates will stay at their present level until at least [date]”) or state-contingent (“rates will stay low until inflation reaches X% and is judged durable”). Its purpose is to shape expectations of future rates today, so that longer-term market rates move even without a current change in the policy rate — it deliberately reduces uncertainty about the future reaction function by pre-committing to it.

The January 2024 statement does the opposite: it says decisions will be taken meeting-by-meeting based on whatever data arrives, with no pre-set path and no commitment to a particular level or duration of restriction beyond what the incoming data justifies. This preserves full optionality for the Governing Council rather than tying its hands.

Why the distinction matters forward guidance works by lowering (or raising) long-run interest rate expectations even before the ECB actually moves — it is itself a monetary policy instrument. A purely data-dependent, non-committal statement carries no such expectational stimulus: markets must continue to guess the path from data releases rather than from a stated commitment, so the statement describes the decision process, not a signal about the future path.

Common confusion to avoid: data-dependence and forward guidance are not opposites in the sense of one being ‘no information’ — the ECB is still telling markets *what it is watching* (underlying inflation dynamics, transmission strength). The key difference is that it is not telling markets *what it will do* in response, which is exactly what forward guidance does.

Mark Allocation — Part (a) (30 marks)

Tier	Marks	What separates this tier
24-30	Correctly identifies the deposit facility as a standing facility setting the floor of the corridor; strong answers also note the deliberately widened 50bp MRO-DFR spread from July 2022 and why.	
14-23	Correct basic definition but no discussion of its role as the effective floor or the current spread context.	
0-13	Vague or partially incorrect description (e.g. confused with MRO or marginal lending facility).	

Mark Allocation — Part (b) (70 marks)

Tier	Marks	What separates this tier
55-70	Clear, correct contrast: forward guidance is a pre-committed signal about the future path (time- or state-contingent) used to shape expectations; the January statement is explicitly non-committal and meeting-by-meeting. Explains why this distinction matters for expectations/transmission.	
35-54	Correctly identifies that the statement avoids commitment but the explanation of what forward guidance actually is/does is incomplete.	
15-34	General sense that the statement is 'cautious' or 'flexible' without a real definition of forward guidance to contrast it against.	
0-14	No meaningful distinction drawn.	

Question A2 Total: 100 Marks

Section A · Question 3

The Equation of Exchange & the Fisher Equation · 2 × 50 marks

A3**100 marks**Explain each of the following:

- (i) The Equation of Exchange.
- (ii) The Fisher Equation.

FULL MODEL ANSWER**(i) The Equation of Exchange — 50 marks**

The Equation of Exchange is the identity $MV = PY$, where M is the money supply, V is the velocity of circulation of money (how many times, on average, each unit of money is used in a transaction over a given period), P is the price level, and Y is real output. It can equivalently be written $MV = PT$ using total transactions (T) rather than output, in Fisher's original formulation.

Symbol	Meaning
M	The money supply (e.g. $M3$)
V	Velocity of circulation — how many times a unit of money changes hands per period
P	The aggregate price level
Y	Real output (real GDP)

As an identity: $MV = PY$ holds by definition — total spending ($M \times V$) must equal total nominal income/output ($P \times Y$), since every transaction has a buyer and a seller.

As a theory (the Quantity Theory of Money): the identity becomes a testable theory once V is assumed to be roughly stable (determined by payment habits/institutions, not by M , P , or Y in the short run) and Y is assumed to be determined independently by real factors (labour, capital, technology). Under those assumptions, a rise in M translates, more or less proportionally, into a rise in P — the classical foundation for the idea that inflation is ultimately a monetary phenomenon (see also Q. B1(b), where this exact identity is applied to the 2015-2018 and 2021-2022 Euro Area episodes).

(ii) The Fisher Equation — 50 marks

The Fisher Equation links the nominal interest rate to the real interest rate and expected inflation: $i = r + \pi^e$ (approximation), or exactly, $(1+i) = (1+r)(1+\pi^e)$.

Symbol	Meaning
i	Nominal interest rate — the rate actually quoted/paid
r	Real interest rate — the return after adjusting for expected inflation
π^e	Expected inflation over the life of the loan/deposit

Why it matters: borrowers and savers ultimately care about the real return, not the nominal one. If expected inflation rises one-for-one with the nominal rate (the ‘Fisher effect’), the real rate is unchanged — this is the assumption behind treating the central bank's real policy rate (used throughout the IS-TR model, Q. A4/A5) as the variable that actually drives spending decisions, not the nominal rate in isolation.

Illustrative check: with $r = 2\%$ and $\pi^e = 3\%$, the exact Fisher equation gives $i = (1.02)(1.03) - 1 = 5.06\%$, very close to the approximation $i = r + \pi^e = 5.00\%$. The approximation is accurate at low-to-moderate inflation but understates i increasingly as inflation rises, since it drops the small cross-term $r \times \pi^e$.

Link between the two: both equations connect the monetary/nominal side of the economy to the real side — the Equation of Exchange links the money supply to nominal spending and (given V , Y) the price level; the Fisher Equation links nominal and real interest rates via inflation expectations. Together they underpin why central banks can influence real activity (via the real rate) using a nominal instrument (the policy rate).

Mark Allocation — Part (i) (50 marks)

Tier	Marks	What separates this tier
40-50	States $MV = PY$ (or $MV = PT$) correctly, defines every term, and explains the identity-vs-theory distinction (i.e. what additional assumptions turn it into the Quantity Theory of Money).	
25-39	States the equation and defines terms but doesn't explain the identity-vs-theory distinction or the QTM link.	
10-24	States the equation with minor errors in definitions, or no discussion of its use.	
0-9	Incorrect or missing equation.	

Mark Allocation — Part (ii) (50 marks)

Tier	Marks	What separates this tier
40-50	States $i = r + \pi^e$ (and ideally the exact form), defines every term correctly, and explains why the real rate is what matters for spending decisions.	
25-39	States the equation and defines terms but limited discussion of its economic significance.	
10-24	States the equation with minor errors, or no discussion of significance.	
0-9	Incorrect or missing equation.	

Question A3 Total: 100 Marks

Section A · Question 4

A Rise in Government Expenditure: The IS-TR Model · 100 marks

A4

100 marks

Use the IS-TR model to examine how an increase in government expenditure could affect interest rates and economic activity in the short run. You should use a diagram to support your answer.

FULL MODEL ANSWER

Setting up the model

The **IS curve** plots combinations of output (Y) and the real interest rate (r) at which planned spending equals output: $Y = C(Y-T) + I(r) + G + NX$. It slopes downward because a higher real interest rate depresses interest-sensitive spending (mainly investment), reducing equilibrium output.

The **TR curve** (the central bank's reaction/Taylor-rule function — see Q. A5) slopes upward: the central bank raises the real interest rate as output rises above trend (to contain the resulting inflationary pressure) and cuts it as output falls below trend.

Step 1 — The direct effect of the rise in G

Government expenditure (G) is a direct, autonomous component of planned spending. A rise in G increases planned spending at every level of the real interest rate, so the entire IS curve shifts right (from IS to IS'). The TR curve is unaffected — the central bank's reaction function itself hasn't changed, only the position of aggregate demand.

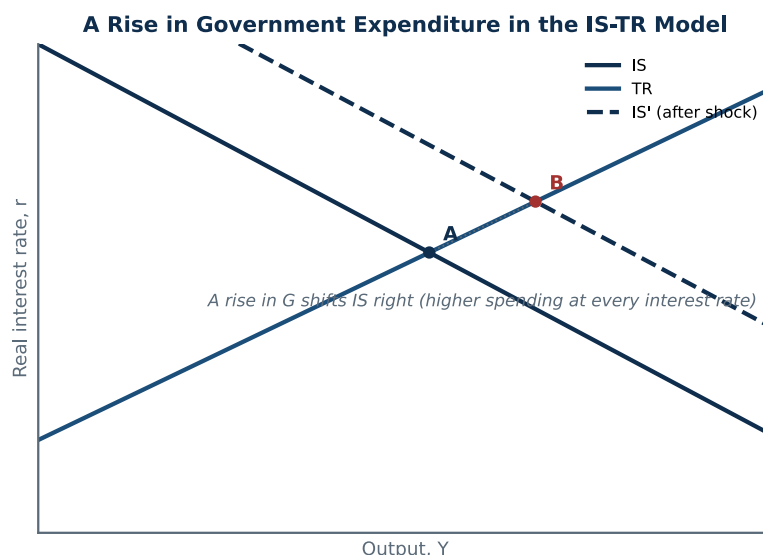


Fig. 1 — A rise in G shifts IS right; the economy moves from A to B along the (unchanged) TR curve

Step 2 — The new short-run equilibrium

The economy moves from the initial equilibrium (A) to a new equilibrium (B), which lies on the original TR curve but on the new, further-right IS' curve. Two things happen simultaneously:

- **Output rises.** At the old real interest rate, planned spending now exceeds output, so firms expand production — output grows (the standard fiscal multiplier effect, before any monetary offset).

- **The real interest rate rises.** Moving along the TR curve, higher output means the central bank now faces stronger demand/inflationary pressure than before, so it responds by raising the real interest rate — this is the central bank's *systematic* response to higher output and higher inflationary pressure, as described by the reaction function itself, not an independent policy shock.

Net effect: output rises, but by less than it would under a fixed interest rate, because the endogenous rise in r (moving up the TR curve) partially offsets the initial expansion by crowding out some interest-sensitive spending (investment). This is the standard **crowding-out** result once the central bank's systematic response is taken into account.

Diagram guidance: one diagram is sufficient — a single IS-TR panel showing the IS curve shifting right with the equilibrium sliding up the unchanged TR curve is all that's required. Compare this directly with the Summer 2025 paper's equivalent question, which asks for a fall in G — the same diagram, mirror image.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
75-100	Correct diagram (IS shifts right, TR unchanged, new equilibrium on the original TR curve) with a full explanation of both the rise in output and the rise in the real interest rate as a movement along TR, and why the interest-rate response partially crowds out the output rise.	
50-74	Correct diagram and correctly identifies that both Y and r rise, but the explanation of <i>why</i> r rises (i.e. the systematic response along TR, not an independent shock) is weak or missing.	
25-49	Identifies that output rises but the model is not properly set up (e.g. TR curve omitted, or IS/TR roles confused) and/or no diagram.	
0-24	Minimal or largely incorrect treatment.	

Question A4 Total: 100 Marks

Section A · Question 5

Interpreting a Central Bank Reaction Function · 100 marks

A5

100 marks

The following equation represents the reaction function of a Central Bank:

$$i = i^n + 1.5(\pi - \pi^T) + 0.5(Y - Y^T)/Y^T$$

where i^n = Natural (Neutral) Rate of Interest, π^T = Inflation Target, Y^T = Trend Output. Explain how the Central Bank changes interest rates, according to this reaction function.

FULL MODEL ANSWER

Reading the equation

The equation says the central bank sets its policy rate (i) equal to a baseline — the neutral rate (i^n), the rate consistent with output at trend and inflation at target — plus two adjustment terms that respond to how far the economy currently sits from that benchmark.

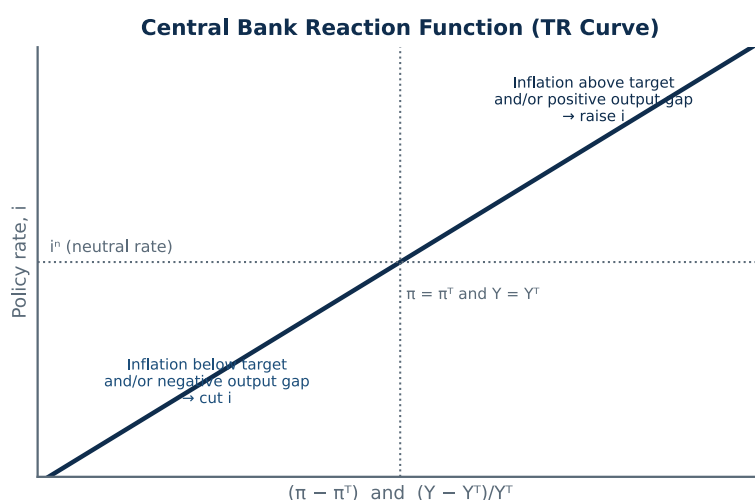


Fig. 1 — The TR curve implied by this reaction function

The two gap terms

Inflation gap, $1.5(\pi - \pi^T)$: if actual inflation is above target, this term is positive and the bank raises the policy rate above neutral. The coefficient of **1.5** means the rate rises more than one-for-one with the inflation gap. Because the real rate is (approximately) $i - \pi$, a more-than-one-for-one nominal response is required to actually raise the real rate and cool demand — a coefficient below 1 would let the real rate fall as inflation rises, which would be destabilising. This is the **Taylor principle**.

Output gap, $0.5(Y - Y^T)/Y^T$: if output is above trend, the term is positive and the bank also raises rates, anticipating the inflationary pressure a positive output gap tends to generate; if output is below trend, the term is negative and the bank cuts rates to support demand. The coefficient of **0.5** is smaller than the inflation coefficient, so the central bank responds less aggressively (in percentage points) to a given percentage output gap than to an equivalent percentage-point inflation gap.

Putting it together

- When $\pi = \pi^T$ and $Y = Y^T$, both bracket terms are zero and $i = i^n$ — the bank simply holds the neutral rate.
- Any deviation of inflation from target, or of output from trend, moves the policy rate away from neutral in the direction that pushes the economy back towards target — the reaction function is inherently stabilising, generating the upward-sloping TR curve used throughout Topic 4 (Q. A4) and Topic 5 (Q. A6, B2).
- If both gaps are positive at once, the two terms reinforce each other and the rate rise is larger than either gap would produce alone.

Illustrative check: with $i^n = 2\%$, $\pi^T = 2\%$: inflation 1pp above target on its own adds 1.5pp to i ($i = 3.5\%$); output 2% above trend on its own adds 1.0pp to i ($i = 3.0\%$); both together add 2.5pp ($i = 4.5\%$). These numbers aren't required for full marks — what matters is correctly explaining the direction and relative size of each term's effect.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
75-100	Correctly explains both gap terms (direction and relative magnitude of the 1.5 vs 0.5 coefficients), explicitly identifies the Taylor principle, and states the baseline case ($i = i^n$ when both gaps are zero).	
50-74	Correctly explains that the bank raises/cuts rates in response to each gap but doesn't discuss why the inflation coefficient needs to exceed 1, or omits the baseline case.	
25-49	General description of 'the bank raises rates when inflation/output is high' without engaging with the structure or coefficients of the equation.	
0-24	Minimal or largely incorrect interpretation of the equation.	

Question A5 Total: 100 Marks

Section A · Question 6

A Positive Energy Shock: The AD/AS Model · 100 marks

A6

100 marks

Use an appropriate model to examine how a positive energy shock (e.g. lower energy prices) could affect economic activity and prices over the medium run. You should use a diagram to support your answer.

FULL MODEL ANSWER

Setting up the model

The appropriate tool is the **AD/AS model** in (output, inflation) space, built from Topic 5. The AD curve (downward sloping) is derived by combining the IS and TR relationships from Topic 4. The AS curve (upward sloping in the short run) reflects firms' price-setting behaviour: at higher output, tighter labour/product markets push costs and prices up; energy prices are a direct input cost that shifts the whole curve.

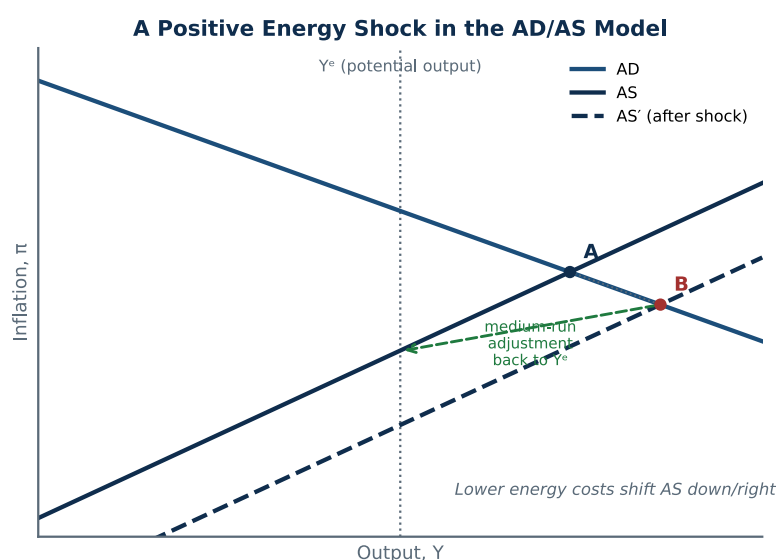


Fig. 1 — A positive energy shock shifts AS right; the economy moves from A to B, then adjusts back towards Y^e over the medium run

Step 1 — The short-run impact

A fall in energy prices directly lowers firms' input costs. At any given level of output, firms can now charge a lower price and still cover costs — this shifts the **AS curve down/right**. The AD curve is unaffected in the first instance (the shock originates on the supply side).

Result: the new short-run equilibrium (B) has lower inflation and higher output than the initial equilibrium (A) — the mirror image of the negative energy shock (stagflation) case. A favourable supply shock moves output and inflation in opposite directions, just like an adverse one, but with both signs flipped: here it's a genuinely welcome outcome, not a policy trade-off.

Step 2 — The medium-run adjustment

Output at B is above potential output (Y^e), which over time puts upward pressure on wages and non-energy costs (via the Phillips-curve/expectations mechanism underlying the AS curve), gradually shifting AS back towards its original position as the temporary cost saving is absorbed into higher demand for labour and other

inputs. In the medium run, absent further shocks, output returns to Y^e and inflation returns towards target, but the price level is permanently lower than it would otherwise have been.

The role of monetary policy: because a favourable supply shock does not put the central bank in a bind the way an adverse one does, the central bank can, if it wishes, allow the temporary boost to output and disinflation to persist a little longer, or use the freed-up policy space (falling inflation with rising output) to ease rates sooner without the trade-off that characterises the negative-shock case (see Q. A6 on the Summer 2025 paper for the adverse-shock mirror image).

Second-round effects: if the lower energy-driven disinflation feeds into lower wage demands and pricing expectations, the AS shift can partially persist rather than fully reversing — but the more common historical pattern (e.g. energy prices easing back in 2023 after the 2022 spike) is that much of the effect does unwind as energy markets normalise, making this a genuinely temporary tailwind rather than a permanent regime change.

Diagram guidance: a single AD/AS panel showing AS shifting right, the short-run equilibrium at B, and an arrow indicating the medium-run drift back towards Y^e is sufficient.

Mark Allocation (100 marks)

Tier	Marks	What separates this tier
75-100	Correct AD/AS diagram with AS shifting right, correctly identifies the short-run outcome (lower inflation, higher output), and explains the medium-run adjustment back towards potential output.	
50-74	Correct diagram and identifies the short-run outcome, but the medium-run adjustment mechanism is underdeveloped or missing.	
25-49	Recognises that prices fall and/or output rises but the model/diagram is not correctly constructed.	
0-24	Minimal or largely incorrect treatment.	

Question A6 Total: 100 Marks

Section B: Long Questions

Answer 1 of 2 · this scheme covers both in full, including both alternatives within B2

Section B questions are broader and typically integrate several ideas together. Question B2 itself offers a further choice: part (a) is compulsory, then you answer either part (b) or part (c). Both alternatives are covered in full below.

Section B · Question 1(a)

What Determines How Much Money Creation Actually Occurs · 3 × 20 marks

B1(a)

60 marks

Examine how each of the following affects money creation:

- (i) Banks’ ability to attract and retain deposits.
- (ii) Business and consumer sentiment in the economy.
- (iii) Liquidity and credit risk facing banks.

In your answer, you should clearly show the relevant transactions on the balance sheets of commercial banks to support your answer.

FULL MODEL ANSWER

Recall from Q. A1 that money is created the moment a bank extends a loan (a new asset and a matching new deposit liability appear together). What determines how much lending — and hence how much money creation — actually takes place is a combination of funding, demand, and risk constraints. Each part below examines one of these constraints.

(i) Banks’ ability to attract and retain deposits — 20 marks

A new loan creates a deposit at the lending bank, but once the borrower spends the funds, that deposit is likely to be paid away to an account at a different bank. The lending bank then needs to fund the resulting outflow of reserves — either from its own existing deposit base, by attracting new deposits, or by borrowing in the interbank/wholesale market or from the central bank.

Lending Bank A	Assets	Liabilities
Step 1: loan drawn down	Loans +100	Customer deposit +100
Step 2: borrower spends, payee banks at Bank B	Reserves −100	Customer deposit −100
Net position after spending	Loans +100, Reserves −100	Deposit 0

Implication: a bank with a strong, sticky deposit franchise can fund more lending cheaply and reliably; a bank that struggles to attract/retain deposits must rely more on interbank borrowing (more expensive, less stable, and itself capped by counterparty limits and liquidity regulation), which constrains how much new lending — and therefore how much money creation — it can sustain.

(ii) Business and consumer sentiment — 20 marks

Unlike (i) and (iii), sentiment operates mainly on the demand side of lending rather than the bank’s own balance sheet constraints.

High confidence: households and firms are more willing to take on debt to finance consumption/investment. Loan applications rise, banks extend more credit (subject to their own capacity), and each new loan creates a new deposit — money creation accelerates.

Low confidence: households and firms retrench, pay down existing debt faster than they take out new debt (echoing the Q. A1 net-repayment scenario), and demand for new lending falls even if banks are willing and able to lend — money creation slows or reverses, independent of any change in banks' own balance sheet capacity.

Balance sheet expression: weak sentiment shows up as loans and deposits both shrinking together, exactly as in the repayment T-account from Q. A1(ii) — the mechanism is identical, only the underlying *cause* (borrower choice rather than a funding constraint) differs.

(iii) Liquidity and credit risk facing banks — 20 marks

Liquidity risk: regulatory liquidity requirements (the Liquidity Coverage Ratio and Net Stable Funding Ratio under Basel III/CRD) require banks to hold enough high-quality liquid assets and stable funding relative to their lending. A bank near these limits must slow new lending (and hence money creation) even if it has willing, creditworthy borrowers, because each new loan uses up liquidity/funding headroom.

Credit risk: banks must hold regulatory capital against risk-weighted assets; riskier loans carry higher risk weights and consume more capital per euro lent. If perceived credit risk rises (e.g. a downturn), banks tighten lending standards and shrink the pool of borrowers they're willing to lend to, directly reducing new loan/deposit creation even for borrowers who would previously have qualified.

Commercial bank balance sheet impact of tighter credit standards	Assets	Liabilities
Loan application declined that would previously have been approved	No change (loan not made)	No change (no deposit created)
Compare: loan that would have been made	Loans +X (foregone)	Deposits +X (foregone)

Net effect: tighter liquidity and credit-risk constraints act as a brake on money creation even when deposit funding (i) is ample and sentiment/demand (ii) is strong — all three constraints can bind independently, and a full answer should treat them as separate, potentially simultaneous limits on the money-creation process, not substitutes for one another.

What ties the three together: (i) and (iii) are constraints on the supply of lending a bank is willing/able to extend; (ii) is a driver of the demand for lending from borrowers. Money creation ultimately requires both a willing lender and a willing borrower — a strong answer keeps this supply/demand distinction explicit rather than treating all three as interchangeable 'reasons lending might be low'.

Mark Allocation — Part (i) (20 marks)

Tier	Marks	What separates this tier
16-20	Correctly shows the balance sheet mechanics of deposit outflow following a loan drawdown and explains the funding constraint this creates.	
8-15	General point that deposits fund lending, without correct balance sheet mechanics.	
0-7	Minimal or incorrect treatment.	

Mark Allocation — Part (ii) (20 marks)

Tier	Marks	What separates this tier
16-20	Correctly identifies sentiment as a demand-side driver, explains both the high- and low-confidence cases, and links back to the balance sheet mechanics of loan creation/repayment.	
8-15	General discussion of confidence affecting borrowing without balance sheet linkage or without covering both directions.	
0-7	Minimal or incorrect treatment.	

Mark Allocation — Part (iii) (20 marks)

Tier	Marks	What separates this tier
16-20	Correctly distinguishes liquidity risk (LCR/NSFR-type constraints) from credit risk (capital/risk-weighting), and explains how each independently constrains new lending/money creation.	
8-15	Mentions liquidity and/or credit risk as constraints on lending without clearly distinguishing the two mechanisms.	
0-7	Minimal or incorrect treatment.	

Section B · Question 1(b)

The Quantity Theory of Money and Recent Euro Area Inflation · 40 marks

B1(b)**40 marks**

“As such the recent experience serves as a reminder that the Quantity Theory of Money is not a vacuous concept of no practical importance for modern day central banks.” (Ref: Thünen Lecture by Isabel Schnabel, Member of the Executive Board of the ECB, Verein für Socialpolitik, Regensburg, 25th Sept. 2023.) Does the Quantity Theory of Money help us to understand the subdued price inflation performance in the 2015-2018 period in the Euro Area and the post-pandemic (2021, 2022) rise in price inflation in the Euro Area? Explain your answer. In your answer, you should use economic theory, your knowledge of how money is created and relevant readings in EC2010 to support your answer.

FULL MODEL ANSWER**The Quantity Theory of Money as the framework**

The Quantity Theory starts from the identity $MV = PY$ (see Q. A3), which becomes a theory once velocity (V) is treated as roughly stable: on that assumption, growth in the money supply should translate fairly directly into growth in nominal spending, and — once real output growth is netted out — into inflation. The Regensburg lecture's argument is that the QTM was too readily dismissed during the low-inflation 2010s, and that the post-pandemic experience shows the underlying mechanism was never dead, only temporarily obscured by velocity behaviour.

2015-2018: subdued inflation despite large asset purchases

What happened to M: the ECB's Asset Purchase Programme (APP), launched in 2015, expanded the monetary base substantially — reserves rose sharply as the Eurosystem bought large quantities of government and corporate bonds.

But inflation stayed subdued: headline and underlying inflation remained persistently below the ECB's target for most of this period, which on a naive reading looks like a QTM failure — large M growth, weak P growth.

The QTM-consistent resolution: velocity fell. much of the new reserve money created did not flow into broad money and spending at the same pace; banks held a large share of the new reserves rather than expanding lending proportionally (weak credit demand after the sovereign debt crisis, deleveraging, a weak money multiplier), and money holdings rose relative to spending (lower velocity) in a low-growth, low-confidence environment — directly echoing the sentiment channel from Q. B1(a)(ii). A fall in V offsetting a rise in M is entirely consistent with the QTM identity.

2021-2022: the post-pandemic inflation surge

Money growth accelerated further and faster: pandemic-era fiscal support (financed in large part by PEPP purchases) pushed broad money growth ($M3$) to historically high rates, this time flowing more directly into household deposits (furlough/support schemes, forced saving during lockdowns) rather than sitting as idle bank reserves.

Velocity rebounded as economies reopened: pent-up demand and the release of accumulated savings once restrictions lifted meant V rose again, reinforcing rather than offsetting the effect of high M growth.

Supply-side shocks compounded the picture: the adverse energy price shock (see the mirror-image question, Q. A6, which asks about a *positive* energy shock) and pandemic-related supply chain disruption pushed AS left at the same time as strong monetary-driven demand pushed AD along a QTM-consistent path — so the 2021-2022 inflation surge reflects both a demand-side QTM story and a genuine supply shock.

Verdict

Partially, but not mechanically

The QTM is useful as an organising framework — it correctly points to money growth as a driver of inflation over a longer horizon, and the 2015-2018 episode is explained, not contradicted, once velocity instability is taken into account. But it is not a reliable short-run forecasting tool on its own: the same monetary expansion produced very different inflation outcomes in 2015-2018 versus 2021-2022 because velocity and the composition of money creation (reserves vs. broad money reaching households) differed sharply between the two episodes, and because supply shocks played an independent role in the second episode.

What a strong answer needs: explicit use of the $MV=PY$ identity, a clear account of *why* velocity moved differently in each period, and integration of the money-creation mechanics from Q. A1/B1(a) rather than treating QTM as a black box.

Mark Allocation (40 marks)

Tier	Marks	What separates this tier
30-40	Correctly sets out $MV=PY$, explains the 2015-2018 episode via a fall in velocity offsetting reserve growth, explains the 2021-2022 episode via faster broad money growth reaching households plus a velocity rebound, discusses the compounding role of supply shocks, and reaches a clear, well-supported verdict.	
20-29	Uses the QTM framework and covers both episodes but the explanation of velocity's role, or the money-creation mechanics behind each episode, is thin.	
10-19	General discussion of inflation drivers in both periods without meaningfully engaging with the Quantity Theory or $MV=PY$.	
0-9	Minimal or off-topic answer.	

Question B1 Total: 100 Marks (60 + 40)

Section B · Question 2(a)

A Policy Rate Increase: IS-TR and AD/AS Together · 60 marks (compulsory within B2)

B2(a)**60 marks**

Use the IS-TR model and the Aggregate Demand/Aggregate Supply model to show how an increase in the policy interest rate will have a contractionary effect on economic activity and reduce price inflation. You should use diagrams to support your answer.

FULL MODEL ANSWER

This question deliberately asks for both models together, because they answer two different halves of the same question: the IS-TR diagram shows *how a policy rate rise reduces output* in the short run; the AD/AS diagram shows *what that fall in output then does to inflation*.

Step 1 — IS-TR: the policy rate rise reduces output

An autonomous increase in the policy rate is modelled as an upward shift of the TR curve: at every level of output, the central bank is now setting a higher real interest rate than before. The IS curve itself is unaffected.

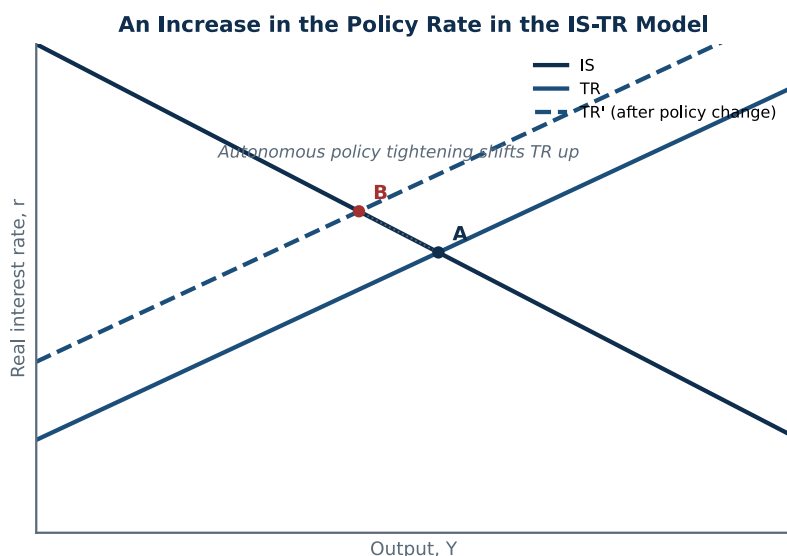


Fig. 1 — A tightening shifts TR up; the economy slides up the (unchanged) IS curve from A to B

Mechanism: the higher real interest rate raises the cost of borrowing and the return to saving, which discourages interest-sensitive spending — principally investment, but also durable consumption. Firms and households cut back, and the economy moves up the IS curve to a new equilibrium (B) with a higher real interest rate and lower output.

Step 2 — AD/AS: lower output feeds through to lower inflation

The AD curve in (output, inflation) space is itself built from the IS-TR relationship. A policy tightening therefore shows up as the AD curve shifting left.

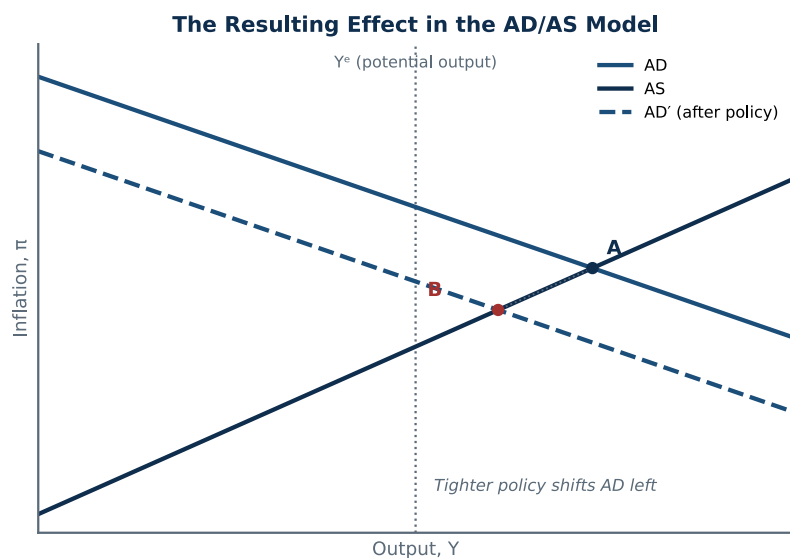


Fig. 2 — AD shifts left along the (unchanged) AS curve; the economy moves from A to B with lower output and lower inflation

Mechanism: with AD to the left and AS unchanged, the new equilibrium (B) lies at both lower output and lower inflation — output and inflation move in the same direction because the shock originates on the demand side.

Putting the two diagrams together

- The IS-TR diagram identifies the *source* of the shock and its *immediate* effect on real activity and the real interest rate.
- The AD/AS diagram then shows the *consequence* for inflation: lower output reduces the pressure on wages and prices, so inflation falls along the AS curve.
- Together: policy rate $\uparrow \rightarrow$ real interest rate $\uparrow \rightarrow$ interest-sensitive spending $\downarrow \rightarrow$ output $\downarrow \rightarrow$ inflation $\downarrow$.

This exact mechanism is the one tested against real data in part (c) below — make sure the two answers are consistent with each other if you're asked both.

Mark Allocation (60 marks)

Tier	Marks	What separates this tier
45-60	Both diagrams correctly drawn (TR shifts up in the IS-TR panel; AD shifts left in the AD/AS panel) with a clear explanation of the transmission chain linking the policy rate rise to lower output and then to lower inflation.	
30-44	Both diagrams present and broadly correct, but the link between the two is not made explicit.	
15-29	Only one of the two required diagrams/models is correctly used, or both are attempted but with significant errors in direction of shift.	
0-14	Minimal or largely incorrect treatment.	

Section B · Question 2(b)

Testing the Model Against the Data · 40 marks (Alternative 1 within B2)

B2(b)**40 marks**

Fig. 1 shows Euro Area Price Inflation and the European Central Bank (ECB) policy interest rate. Price inflation remained well above target despite the ECB increasing the policy rate by 400 bps, from -0.5% in Sept. 2019 to 3.5% in June 2023. The ECB raised the policy rate further to 4% in Sept. 2023. Does this mean that the models in (a) have little to tell us about how higher interest rates can affect price inflation? Explain your answer. In your answer, you should draw on your economic theory in EC2010 and your understanding of the dynamics of price inflation today to support your answer.

And Either

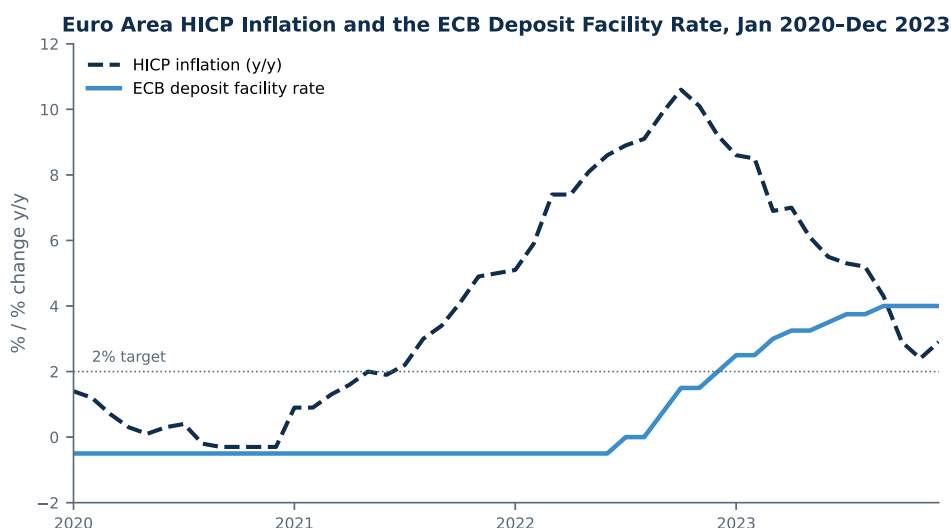
FULL MODEL ANSWER

Fig. 1 — Illustrative reconstruction of the published path of Euro Area HICP inflation and the ECB deposit facility rate, based on Eurostat/ECB data

No — the models still apply; several factors explain the apparent puzzle

The models in (a) are not undermined by this pattern

1. Monetary policy works with long, variable lags. the IS-TR/AD-AS transmission chain in part (a) takes time to play out — typically estimated at 12–24 months from a rate change to its peak effect on inflation. Rates only started rising materially from July 2022; by the time of writing (early-to-mid 2024) much of that tightening had not yet fully worked through the system. Inflation falling from its October 2022 peak of over 10% to under 3% by late 2023 while rates were still being raised is consistent with the model working, with a lag, rather than not working at all.

2. This episode began as a large adverse supply shock, not a demand shock. the initial surge in inflation (2021-2022) was driven substantially by energy prices and pandemic-related supply chain disruption — exactly the AS-curve shift examined in Q. A6's mirror-image question. The AD/AS model itself predicts that raising rates in response to a *supply* shock is less effective at controlling inflation, and comes at a higher output cost, than raising rates in response to a genuine demand shock (see the policy trade-off discussed in the Summer 2025 paper's version of Q. A6). Much of the 2022-2023 disinflation reflects energy prices falling back (a supply-side reversal) rather than the ECB's tightening alone.

3. The starting point matters — real rates were deeply negative. with inflation running above 8-10% for much of 2022 and nominal rates still below that for most of the tightening cycle, the real policy rate ($i - \pi^e$, see the Fisher Equation, Q. A3(ii)) remained negative or only mildly positive for a long stretch even as the nominal rate rose sharply. It is the real rate, not the nominal rate, that actually restrains spending in the IS-TR model — so a large nominal hike does not necessarily translate into an equally large tightening in the real rate people and firms actually face.

4. The policy did eventually work. by late 2023 inflation had fallen substantially from its peak while output growth slowed, which is exactly the AD/AS prediction once (i) the lags have had time to operate and (ii) the initial supply shock itself faded. The chart shows disinflation accelerating through 2023 as the cumulative effect of the tightening cycle builds — not a case of rates having no effect, but of effects taking time.

Verdict

The apparent puzzle — large rate hikes alongside persistently high inflation for over a year — does not mean the IS-TR/AD-AS framework in (a) is wrong. It reflects (i) transmission lags the model itself predicts, (ii) the fact that this was substantially a supply-driven inflation episode which the model also predicts is harder and costlier to control via rates alone, and (iii) the real interest rate (the variable that actually matters in the model) rising by much less than the nominal rate during the period when inflation was highest. Once these are accounted for, the data are broadly consistent with the model, not evidence against it.

What a strong answer needs: don't just describe the chart — use it as evidence *for or against* a specific claim about the models, engaging with lags, the supply-shock origin of the inflation, and the real-vs-nominal rate distinction. A purely descriptive answer with no engagement with the theory scores poorly here.

Mark Allocation (40 marks)

Tier	Marks	What separates this tier
30-40	Engages substantively with at least three of: transmission lags, the supply-shock origin of the inflation (and why that changes the model's own prediction), the real-vs-nominal rate distinction, and the observed disinflation through 2023 as evidence the mechanism did operate. Reaches a clear, well-argued verdict rather than just describing the chart.	
20-29	Correctly argues the models aren't undermined but only develops one or two of the above points in depth.	
10-19	Mostly descriptive account of the chart with limited engagement with the theoretical models from (a).	
0-9	Minimal or off-topic answer.	

Question B2 (with option b) Total: 100 Marks (60 + 40)

Section B · Question 2(c)

Projecting the ECB Policy Rate for H2 2024 · 40 marks (Alternative 2 within B2)

B2(c)**40 marks***Or*

Provide your projection for the ECB's policy interest rate for the second half of 2024. In your answer you should state clearly the assumptions that you are making in order to derive your answer.

FULL MODEL ANSWER

This is an open forecasting question — there is no single "correct" number, and the mark scheme rewards a well-reasoned projection built transparently on stated assumptions using the course's own framework (the TR/reaction-function logic from Q. A5), not any particular numerical answer. A strong answer as of Summer 2024 might have looked like this.

Step 1 — State the starting point and the assumptions

- **Starting point:** deposit facility rate at 4.00% (unchanged since Sept 2023, per Q. A2).
- **Assumption 1 — disinflation continues but slows:** headline HICP inflation has fallen sharply from its 2022 peak and is approaching the 2% target, but underlying/core inflation (particularly services) remains stickier, reflecting continued wage growth as labour markets stay tight.
- **Assumption 2 — growth is weak but not collapsing:** Euro Area growth has been sluggish, reducing the urgency for further tightening but not (on this assumption) triggering an emergency-style rapid cutting cycle.
- **Assumption 3 — the ECB behaves broadly as its own reaction function (Q. A5) predicts:** as both the inflation gap and (to a lesser extent) the output gap narrow, the TR curve implies the policy rate should move back down towards its neutral level, gradually rather than abruptly, consistent with the data-dependent, meeting-by-meeting approach described in Q. A2(b).

Step 2 — Apply the reaction function logic

Using the same reaction function as Q. A5, $i = i^n + 1.5(\pi - \pi^T) + 0.5(Y - Y^T)/Y^T$: if the inflation gap is assumed to narrow steadily over H2 2024 (from a still mildly-elevated exit rate towards target) while the output gap stays modestly negative, the reaction function itself implies a declining path for i over the second half of the year — the direction of travel matters more here than the precise numbers.

Step 3 — State the projection

Projection: a gradual, data-dependent easing cycle beginning around mid-2024, most plausibly with the first 25bp cut once the Governing Council judges underlying disinflation to be sufficiently assured (consistent with the January 2024 statement's emphasis on 'the dynamics of underlying inflation' and 'the strength of monetary policy transmission', Q. A2(b)), followed by further gradual cuts through H2 2024 if disinflation continues on track — taking the deposit facility rate from 4.00% down by somewhere in the region of 50–100bp by the end of 2024, delivered in small (25bp) steps consistent with the gradualism/Brainard-conservatism logic (uncertainty argues against large moves in either direction).

Key risks to the projection stated explicitly: a resurgence in energy prices, stronger-than-expected wage growth feeding into services inflation, or a shock to inflation expectations could all justify holding rates higher for longer than this baseline path; conversely, a sharper-than-expected growth slowdown could justify faster or larger cuts.

What actually happened (for study reference — this postdates the exam and is not required for full marks): the ECB delivered its first cut of this cycle in June 2024 (deposit facility rate 4.00% → 3.75%), followed by further 25bp cuts in September, October, and December 2024, taking the rate to 3.00% by year-end — broadly in line with the gradual, meeting-by-meeting easing path a well-reasoned answer along these lines would have projected, though the pace (four cuts rather than two to three) came in a little faster than the illustrative 50-100bp range above.

Mark Allocation (40 marks)

Tier	Marks	What separates this tier
30-40	States clear, economically sound assumptions (inflation trajectory, output gap, ECB reaction-function behaviour), applies the reaction-function logic from Q. A5 correctly to derive a directionally sound projection (rates gradually easing, delivered in small steps), and explicitly discusses risks to the projection in both directions.	
20-29	Provides a reasonable projection with some stated assumptions, but the link to the reaction-function framework is weak or the direction of travel is not well justified.	
10-19	States a number with minimal justification or assumptions.	
0-9	No meaningful projection or assumptions given.	

Marking note: since this is an open forecasting question, a well-reasoned answer projecting rates *on hold* through H2 2024 (e.g. on the assumption that underlying inflation proves stickier than expected) should score just as highly as one projecting cuts, provided the assumptions are stated clearly and applied consistently via the course's own models. Do not penalise a directionally different answer that is well argued — penalise only unsupported assertions.

Question B2 (with option c) Total: 100 Marks (60 + 40)